MILITARY SPECIFICATION SHEET

TUBING, PLASTIC, EXTRA FLEXIBLE, CONVOLUTED, POLYTETRAFLUOROETHYLENE CLOSE CONVOLUTION

This specification has been approved by the Naval Air Systems Command, Department of the Navy

The complete requirements for procuring the tubing described herein shall consist of this document and the issue in effect of MIL-T-81914 (AS).

REQUIREMENTS:

Convolution type: Helical (see 3.3)

Construction details: Construction details, including available sizes, dimensions and tolerances are located in Table I. Lengths shall be as specified by the procuring activity.

Continuous operating temperature: -67°C (-88°F) to +260°C (500°F)

Color: Unless otherwise specified, the supplied color shall be black.

Physical properties: General physical requirement values, along with associated test conditions are located in Table II.

Military Part Number: Consists of the basic number of this specification sheet and dash numbers as shown below:

M81914/2-0000

- Identifier (Table I)
- Class, when applicable (when no classes are specified, the numeric 0 shall be used)
- Type of convolution (1- helical, 2- annular)
- Specification sheet

Close convolution polytetrafluoroethylene, thin wall, 0.500 inch is identified as: M81914/2-1006

FSC 9330

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TABLE I

CONSTRUCTION DETAILS

Identifier	Max. Inside Dia.	Min. Inside Dia.	Max. Outer Dia.	Wall Thickness Max.	Conv. Per Inch ± 1	Weight (Lbs) Per 100 feet Max.	Min. Bend Radius
**01	.188	.181	.320	.023	10	2.2	.313
**02	.281	.273	.414	.026	9	3.8	.438
**03	.312	.306	.450	.02	9	4.8	.438
**04	.375	.364	.530	.029	9	5.6	.500
**05	.437	.427	.590	.029	9	6.5	.500
**06	.500	.485	.660	.029	9	6.9	.750
**07	.625	.608	.780	.029	9	9.2	.750
**08	.750	.730	.975	.035	8	13.8	.938
**09	.875	.860	1.100	.035	8	15.0	.938
**10	1.000	.975	1.260	.035	7	16.8	1.125
**11	1.125	1.105	1.390	.035	6	17.5	1.125
**12	1.250	1.210	1.539	.035	6	19.6	1.250
**13	1.500	1.450	1.810	.038	6	26.0	2.00

** - The asterisks shall be replaced by Convolution Type and Class designation.

Note: Unless otherwise specified all dimensions are in inches.

TABLE II

PHYSICAL PROPERTIES

PROPERTY	REQUIREMENT	TEST METHOD
Construction details	In accordance with Table I	4.6.1
Stress in psi @ 10% strain	150 to 450	4.6.2
Specific gravity, max.	2.20	4.6.3
Crush resistance, pounds horizontal, min.	10	4.6.4
Low temperature flexibility	No cracking	4.6.5; 36,000 cycles

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TABLE II (Continued)

PROPERTY	REQUIREMENT	TEST METHOD
Heat shock	No flowing, dripping or cracking	4.6.6; 232 ±2°C (450 ±5°F) for 4 hours
Longitudinal change, %, max.	15	
Heat aging		
Stress in psi @ 10% strain; % min. of as rec'd value	80	4.6.7; 232 ±2°C (450 ±5°F) for 96 hours
Fluid resistance		4.6.8
Stress in psi @ 10% strain; % min. of as rec'd value	80	
Flammability	Non-flammable 1/	4.6.9
Fungus resistance	Non-nutrient 1/	4.6.10
Dielectric breakdown	No breakdown, 12,000 volts	4.6.11

1/ See paragraph 6.4; Properties not tested.

Standard procurement: Tubing shall be procured as coils. Unless otherwise specified, individual lengths shall be 18 feet.

Preparing Activity
Navy - AS
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-T-81914/2(AS) TUBING, PLASTIC, EXTRA FLEXIBLE, CONVOLUTED, POLYTETRAFLUOROETHYLENE CLOSE CONVOLUTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

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